

Hexuan (Matt) Wang

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Research Interests

As large models move into high-stakes settings—scientific discovery, real-time decision-making—being **faithful and verifiable** matters as much as being capable. My research treats **faithfulness as a property that should hold across modalities**: every claim a model makes, and every output it generates, should be **traceable to its evidence and open to human verification**. I pursue this on two complementary fronts. First, I *characterize when and why* models are faithful—through rigorous behavioral evaluation, using probing only as a diagnostic to surface the gap between what a model internally registers and what it states aloud. Second, I *ground* reasoning and generation in increasingly complex settings, from citation attribution in text, to reasoning under temporal shift, to traceable multimodal generation. My interests are inherently *cross-modal*: faithfulness should not stop at text, but become a unifying design principle.

Education

Johns Hopkins University

Aug 2023 – May 2027 (Expected)

B.S. in Computer Science, Applied Mathematics & Statistics

GPA: 3.98 / 4.00

- **Graduate-Level Coursework (taken as an undergraduate)**: Advanced Self-Supervised Statistical Models, Machine Social Intelligence, Matrix Theory.
- **Selected Coursework**: Natural Language Processing, Probabilistic Models of the Visual Cortex, Machine Learning, Mathematical Image Analysis, Mathematical Statistics & Probability.

Publications & Preprints

Bold = author; * = equal contribution.

- [1] **Hexuan Wang**, Yaxuan Ren, Srikar Bommireddypalli, Shuxian Chen, Adarsh Prabhudesai, Rongkun Zhou, Elina Baral, Philipp Koehn. “SciTaRC: Benchmarking QA on Scientific Tabular Data that Requires Language Reasoning and Complex Computation.” *Accepted at COLM 2026; earlier version at SURGeLLM Workshop @ ACL 2026*. [arXiv:2603.08910](https://arxiv.org/abs/2603.08910).
- [2] **Hexuan Wang**, Jingyu Zhang, Benjamin Van Durme, Daniel Khashabi. “Are Finer Citations Always Better? Rethinking Granularity for Attributed Generation.” *Under review at TMLR*. [arXiv:2604.01432](https://arxiv.org/abs/2604.01432).
- [3] **Hexuan Wang**, Jaemin Cho. “Bidirectional SVG–Image Attribution for Grounded Image Editing.” *In preparation (targeting ICLR 2027)*.
- [4] **Hexuan Wang**, Christian Rispoli, Tianjian Li, Jingyu Zhang, Xi Wang, Benjamin Van Durme, Fartash Faghri, Mehrdad Farajtabar, Eric Nalisnick, Daniel Khashabi. “Training in the Past, Reasoning about the Future: Principled Evaluation of Temporal Generalization.” *In preparation (targeting ICLR 2027)*.

Talks & Posters

- **Hexuan Wang**, Daniel Naiman. “Spectral Analysis and Mathematical Reconstruction of FM Synthesis.” Selected Talk, MATRX; Poster, SIAM DMV Applied Mathematics Conference, 2025.

Research Experience

Faithful LLM Reasoning: Evaluation, Verification & Training

May 2025 – Present

Advisor: Prof. Daniel Khashabi (Computer Science, CLSP)

Johns Hopkins University

- **Citation Granularity.** Led an empirical study showing that enforcing fine-grained (sentence-level) citations *degrades* attribution quality, while intermediate (paragraph-level) granularity optimizes both verifiability and correctness.
- **Logic Probing.** Found a metacognitive blind spot in reasoning models—internally they register that information is missing (logical underspecification), yet confidently hallucinate aloud; used linear probes *purely as a diagnostic* to detect this signal, which in turn makes it possible to prompt the model to ask a clarifying question rather than make silent assumptions.
- **Chrononauts.** Built a frontier-scale, contamination-aware pipeline for temporal generalization—enforcing strict temporal filtering across continual pretraining (context extension) and post-training (SFT + RL) to cleanly separate knowledge acquired in pretraining from reasoning expressed at inference; released the first suite of temporally-controlled aligned models, evaluated across temporal QA, forecasting, and scientific reasoning.

Grounded Image Editing via Bidirectional Attribution

Mar 2026 – Present

Advisor: Prof. Jaemin Cho (Computer Science)

Johns Hopkins University

- Training multimodal LLMs to align **SVG code** and **rendered images** through *bidirectional attribution*, so that edits are grounded in—and traceable to—a symbolic, verifiable representation of the image.

Scientific Tabular Reasoning & Neuro-Symbolic Evaluation

Dec 2024 – Jan 2026

Advisor: Prof. Philipp Koehn (Computer Science, CLSP)

Johns Hopkins University

- **SciTaRC benchmark.** Built an expert-authored scientific tabular-QA benchmark requiring deep language reasoning and computation, augmented with pseudo-code plans and complexity metrics.
- **Bottleneck analysis.** Built an LLM evaluator with 97% human alignment; benchmarked frontier models to reveal a universal “execution bottleneck”—failures stem from poor execution fidelity rather than strategy formation.

Spectral Analysis & Reconstruction of FM Synthesis

Nov 2023 – Sep 2024

Advisor: Prof. Daniel Naiman, Dept. of Applied Mathematics & Statistics

Johns Hopkins University

- Built a Python/PyTorch FM synthesizer by reconstructing core C++ algorithms; used Short-Time Fourier Transform to analyze modulation envelopes and design an audio reconstruction algorithm from spectral components.

Baseball Scheduling Optimization

Jan 2024 – Feb 2025

Advisor: Prof. Anton Dahbura, Information Security Institute

Johns Hopkins University

- Formulated large-scale integer linear programs and combinatorial optimization algorithms (Gurobi, Hexaly), delivering scheduling solutions adopted league-wide by Major and Minor League Baseball (MLB & MiLB).

Honors & Awards

- **Pistritto Undergraduate Research Fellowship** (\$6,000), JHU Computer Science — competitive, faculty-committee-selected award for undergraduate research. 2026
- **Naddor Prize**, Dept. of Applied Mathematics & Statistics, JHU — for distinguished academic performance by a non-senior AMS student. 2026
- **Dean’s List**, Johns Hopkins University Every semester

Teaching

Teaching Assistant

Johns Hopkins University

Jan 2024 – Present

Baltimore, MD

- **Graduate courses:** Artificial Intelligence; Numerical Linear Algebra.
- **Undergraduate courses:** Mathematical Foundations for Computer Science; Linear Algebra for Data Science; Linear Algebra & Differential Equations.
- Led weekly recitations and office hours; graded 160+ exams/assignments weekly; designed advanced algorithmic and mathematical practice materials.

Technical Skills

- **LLM/MLLM Training & Post-training:** continual pretraining (context extension), supervised fine-tuning (SFT), reinforcement learning (RLHF/RLVR, DPO), parameter-efficient fine-tuning (LoRA/PEFT); PyTorch, HuggingFace (Transformers/TRL/PEFT), OpenRLHF, DeepSpeed/FSDP, vLLM, Accelerate.
- **Multimodal (MLLM):** training & evaluation of multimodal LLMs (MLLM/VLM), image generation & editing, vision–language alignment and attribution.
- **Evaluation & Methods:** LLM/MLLM evaluation, attribution & retrieval-augmentation (RAG), causal inference (graphical models, missing data), statistical learning, combinatorial optimization (Gurobi, Hexaly), signal processing (FFT, STFT).
- **Languages & Tools:** Python, MATLAB, SQL, Java, C/C++, $\LaTeX$, Mathematica, Git.

Leadership & Service

JHU Vocal Chords A Cappella

Co-President, Business Manager & Bass

2023 – Present

Johns Hopkins University

- 3rd Place, ICCA Quarterfinals (2024, 2026); managed a \$3,000+ production and equipment budget.